
SCCM OSD & Imaging – Process, Q&A, and Troubleshooting

◆ OSD & Imaging Process (High-Level Flow)

1. Prepare Environment

- Configure **Boot Images (x64/x86 WinPE)** with network/storage drivers.
- Enable PXE on DP (WDS role or CMG/Cloud PXE).

2. Create Task Sequence (TS)

- Capture image (sysprep → capture media) or use default WIM.
- TS includes **Format/Partition → Apply OS Image → Apply Drivers → Apply Updates → Install Apps → ConfigMgr client install → Join Domain → Customizations**.

3. Deploy Task Sequence

- TS deployed as **Available** (boot media) or **Required** (zero-touch).
- Target device collection.

4. Boot Process

- PXE request → DP/WDS responds with boot image.
- WinPE loads and contacts MP.

5. Execution

- TS runs step by step.
- Status messages sent to MP/Site server.

6. Completion

- Device reboots into full OS with SCCM client.
- Applications, updates, and policies enforced.

◆ Interview Questions & Answers (Scenario-Based)

Q1. Explain SCCM OSD architecture.

A: SCCM OSD uses **task sequences** to automate imaging. Boot images load WinPE, contact

the MP, and download policies. The TS applies WIM image, installs drivers, apps, and joins domain. PXE-enabled DPs or boot media are used. Communication flows through **MP → DP → Client**.

Q2. What logs are critical during OSD?

A:

- **smspxe.log** (PXE boot issues, DP logs)
 - **smsts.log** (main OSD troubleshooting, present in X:\Windows\Temp\SMSTSLog during WinPE)
 - **distmgr.log** (package distribution issues)
 - **mpcontrol.log** (management point connectivity)
 - **drivercatalog.log** (driver import)
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Q3. Device not PXE booting – how do you troubleshoot?

A:

- Check DHCP scope options (66 = DP, 67 = boot\x64\wdsnbp.com).
 - Validate **PXE enabled on DP** and WDS service running.
 - Review **smspxe.log** on DP for requests/errors.
 - Ensure device MAC/UUID not in **Unknown computer support conflicts**.
 - Clear PXE cache entry in SCCM console if needed.
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Q4. Task Sequence fails at “Applying Operating System” – steps?

A:

- Review **smsts.log** for error code (often 0x80070002 file not found).
 - Check if OS image package is distributed to DP.
 - Validate network access in WinPE (press F8 → ipconfig, ping DP).
 - Confirm correct WIM file path in TS.
 - Redistribute package if corrupt.
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Q5. Drivers not installing during OSD – why?

A:

- Boot image missing NIC/storage drivers → check with F8 → ipconfig.
 - Wrong driver package linked in TS.
 - Outdated driver catalog or incorrect Plug and Play ID.
 - Fix by updating boot image with drivers and re-distributing.
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Q6. TS fails at domain join step. What do you check?

A:

- Verify account used in TS has **domain join permissions**.
 - Check DNS resolution in WinPE/full OS.
 - Ensure network profile is applied correctly.
 - Review **netsetup.log** on client.
 - Validate OU path in TS is correct.
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Q7. Application not installing during OSD. Troubleshooting?

A:

- Check if app is deployed to **device collection**.
 - Confirm app content is distributed to DP.
 - Review **AppEnforce.log** post-OS deployment.
 - Test manual installation on client.
 - Use TS retry step for failed installs.
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Q8. How do you handle driver management in SCCM OSD?

A:

- Use **driver packages** per model.
- Inject critical drivers (NIC/Storage) into boot images.
- Apply drivers dynamically using **WMI query** in TS (e.g., SELECT * FROM Win32_ComputerSystem WHERE Model LIKE "%Latitude 7420%").

- Keep drivers updated regularly.
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Q9. TS runs fine but computer stuck in reboot loop after deployment.

A:

- Usually due to **boot image assigned as OS** instead of WIM.
 - Check **Apply Operating System step** references correct WIM.
 - Ensure boot media is not corrupted.
 - Review smsts.log for step execution.
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Q10. How do you troubleshoot slow OSD deployments?

A:

- Check **network bandwidth** between DP and client.
 - Validate DP performance and package distribution.
 - Enable **multicast** for large-scale deployments.
 - Split TS into phases and optimize drivers.
 - Monitor logs: smspxe.log (PXE) and smsts.log (TS progress).
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◆ Troubleshooting Guide (Quick Reference)

Scenario	Logs to Check	Key Steps
PXE boot fails	smspxe.log, WDS logs	Check DHCP, DP PXE settings, MAC conflicts
WinPE boots but no TS	smsts.log, MP logs	Ensure TS deployed to collection, check boundaries
OS Image step fails	smsts.log	Validate WIM on DP, network in WinPE
Drivers missing	smsts.log, setupact.log	Add drivers to boot image/driver package
Domain join fails	netsetup.log, smsts.log	Check permissions, OU path, DNS
Apps not installing	AppEnforce.log, smsts.log	Redistribute content, check dependencies
OSD very slow	smsts.log, distmgr.log	DP performance, package distribution, use multicast
Client not registering	ccmsetup.log, clientidmanagerstartup.log	Verify MP connectivity, reinstall client

1) Common prerequisites (applies to Imaging & In-Place Upgrade)

1. SCCM/MECM infra: Site server, Management Point, Distribution Points (DP) and PXE-enabled DPs where needed. Ensure DPs have enough disk and are in the correct boundary groups.
 2. SUP (Software Update Point) and WSUS for patching if you plan to deploy updates in TS. Ensure SUP syncs languages/metadata you need.
 3. Network: DHCP + PXE (or IP helpers) configured; firewall allows SMB/HTTPS to DPs/MPs. If using CMG/Cloud PXE, ensure cloud services configured.
 4. Accounts: Domain join account with permission to create computer objects (or prestage) and an account for content distribution. Document all passwords and service accounts securely.
 5. Source files: Clean reference Windows WIMs or Windows setup files for an upgrade package. Have drivers, apps, and packages created and distributed to DPs ahead of the deployment.
 6. Test lab: Always validate on VMs or a pilot device collection first (different hardware models, UEFI/BIOS, SSD/HDD).
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2) Imaging (Bare-metal / Wipe & Load) — Step by step

A. Prepare reference image (Build & Capture)

1. **Create a clean reference machine (VM or hardware)**
 - Install Windows (target edition and build), apply latest cumulative updates and language packs. Do not join domain (or join and then remove). Install only required apps for the reference image (usually minimal).
 - Configure OS settings and desired registry tweaks, disable telemetry if required by org policy.
2. **Install prerequisites for image capture**
 - Install SCCM client (if using capture TS this can be automatic). Remove machine-specific items (machine SID issues handled by sysprep). Clean up temp files and run DISM /Cleanup-Image if required.

3. Sysprep & Generalize

- Use Sysprep with **Generalize + OOBE** (unless you want to keep certain settings). Best practice: let SCCM Build and Capture TS handle sysprep automatically. If doing manual sysprep: sysprep /generalize /oobe /shutdown (then boot capture media).

4. Create Capture Task Sequence

- Use SCCM console → Software Library → Operating Systems → Task Sequences → **Create Task Sequence** → **Capture a reference operating system image**.
- Steps the TS will run: Prepare, Sysprep, Capture WIM (store on network share or DP). Provide UNC share for capture (\\server\captures\$) and credentials.
- Distribute the capture package and the boot image to the DP(s) that will PXE boot the target.

5. Boot into WinPE and capture

- PXE boot the reference device to WinPE (or use capture media). The TS will sysprep and capture the WIM and place it on the network share (or DP). Verify captured WIM integrity.

B. Prepare boot images & drivers

1. Create/Update Boot Images (WinPE)

- In SCCM, update default x86/x64 boot images as needed. Add network/NIC and storage drivers to the appropriate boot images so WinPE can access the network and DP. Then redistribute updated boot images to DPs.

2. Driver Strategy

- Inject only critical drivers (NIC / storage) into the boot image. For model-specific drivers use **driver packages** per model (do not import every OEM driver into the boot image). Use WMI queries (model/manufacturer) in TS to apply the correct driver package.

C. Create Deploy Task Sequence (Wipe & Load)

1. Create Task Sequence

- Console → Software Library → Operating Systems → Task Sequences → **Create Task Sequence** → **Deploy an existing image package** or **Create a new custom TS**. Typical TS steps in order:

1. **Format and Partition Disk** (UEFI GPT or BIOS MBR)

2. **Apply Operating System Image** → select WIM and index
3. **Apply Windows Settings** (product key, local admin)
4. **Apply Network Settings / Join Domain** (or separate step later)
5. **Apply Driver Package** (model-targeted WMI query)
6. **Apply Windows Updates** (Install Software Updates)
7. **Install Applications** (Install Application steps)
8. **Run Scripts / Configure** (Run Command Line steps)
9. **Setup Windows and ConfigMgr** (install SCCM client)
10. **Restart & Finalize** → domain join, run compliance, remove any temporary accounts.

2. Partition Layout recommendations

- UEFI (GPT): EFI 100 MB FAT32, MSR 16 MB, Primary (C:) rest NTFS/NTFS compression off, Recovery ~450 MB (or vendor recommended).
- BIOS (MBR): System reserved 100 MB, Primary (C:) rest NTFS, Recovery partition optional. Use the Format and Partition Disk step to create exact layout.

D. Distribute & Deploy

1. **Distribute OS image (WIM) & Driver packages** to DP(s) used for PXE or PXE content. Use Distribution Points and ensure content status shows success.
2. **Enable PXE on selected DP(s)**: Configure PXE responders, WDS (on-prem), ensure smspxe.log shows requests. If using IP helpers, ensure network team has configured them.
3. **Deploy TS to Device Collection**: Typically Required for zero-touch or Available for lite-touch. Configure PXE and media usage.
4. **Boot target device to WinPE (PXE or USB)** and select the TS. Monitor via smsts.log.

E. Post-deployment validation

1. Verify device shows up in SCCM console and that the client is active. Check **ccmsetup.log**, **clientidmanagerstartup.log**.
2. Validate application install success via **AppEnforce.log**, ensure Windows updates applied and drivers installed properly.
3. Remove temporary accounts, verify group policies and OU settings.

3) In-Place Upgrade (Feature Upgrade) — Step by step

A. Prerequisites & planning

1. **Supported upgrade path:** Confirm source OS → destination OS is supported (e.g., Windows 10 21H1 → Windows 10 22H2 or Windows 10 → Windows 11 only where hardware supports). Check application compatibility.
2. **Driver compatibility:** Gather and test latest OEM drivers. Driver issues are the most common cause of upgrade rollbacks (error 0xC1900101).
3. **Disk space & free space checks:** Ensure device has required temporary space (~20 GB recommended for feature upgrades).
4. **BitLocker:** Suspend BitLocker before upgrade (or pre-provision encryption). Use `manage-bde -protectors -disable C:` or BitLocker TS steps. Plan to re-enable after upgrade.
5. **User State:** If user data needs explicit protection, use USMT (scanstate/loadstate) — though in-place upgrade usually retains profiles; USMT is used when you want a safe rollback or migration between editions.

B. Build the upgrade package

1. **Create an Upgrade Package:** In SCCM console create an **Operating System Upgrade Package** using `setup.exe` and the `install.wim` (or use media from Microsoft). Distribute package to DP(s).
2. **Pre-cache content:** Use Pre-cache TS or Schedule pre-download to local cache (Delivery Optimization or Peer Cache) to reduce network load during rollout.

C. Create the Upgrade Task Sequence

1. **Create Task Sequence - Upgrade the operating system** using the SCCM wizard (Operating Systems → Upgrade Operating System). The TS runs in the full Windows environment (not WinPE).
2. **Typical TS steps order:**
 1. **Preflight checks** (custom script: check free space, check pending reboot, check incompatible apps)
 2. **Suspend BitLocker** (if applicable)
 3. **Run Compatibility Scan** (script or `Setup /compat`)
 4. **Upgrade Operating System** → point to the Upgrade Package (`setup.exe`)

5. **Install drivers / Driver updates** (post-upgrade)
6. **Install applications/updates** (Install Application & Install Software Updates)
7. **Re-enable BitLocker** (resume protectors)
8. **Repair/Reinstall ConfigMgr client** if required (ccmsetup with /mp: switch)
9. **Post-upgrade checks & health scripts**, send compliance status.

D. Deployment & execution

1. **Pilot group:** Always pilot on small collections by hardware type and user profile. Monitor time to upgrade, failures, and UX impact.
2. **Monitor progress:** Use SCCM deployment monitoring and Client logs. For in-place upgrades, Windows Setup logs are critical (Panther folder).
3. **Rollback:** If Windows Setup fails mid-upgrade, it will attempt automatic rollback to previous OS. After rollback, collect Panther logs and TS logs to root cause. If rollback completed, SLA the device for reattempt or manual remediation.

E. Post-upgrade validation

1. Confirm OS version, driver versions, installed critical apps and windows updates. Check event viewer and run **sfc /scannow** or **DISM** if corruption suspected.
2. Verify SCCM client registration and policy retrieval (ccmexec service running). Reinstall client if it was damaged.
3. Re-enable BitLocker and verify TPM and encryption state.

4) USMT (User State Migration) notes — when to use & steps

1. **When to use:** Use USMT when you must preserve user profiles across wipe & load imaging or when migrating users between machines/editions during a controlled migration. For in-place upgrades USMT is optional because Windows setup preserves profiles by default.
2. **ScanState (capture):** scanstate \\server\store\USMTStore /i:migapp.xml /i:migdocs.xml /o /c /v:13 /uel:90 — run in WinPE or full OS depending on TS design. Store on network share or DP.
3. **LoadState (restore):** loadstate \\server\store\USMTStore /i:migapp.xml /i:migdocs.xml /v:13 — run post-deploy in TS.
4. **Encryption & security:** Use /encrypt or share secured with ACL because store contains user data. Use migration XMLs carefully to include/exclude data.

5) Task Sequence: detailed recommended step list (Imaging TS example)

1. **Prestart / Validation:** Check variables (SMSTSPackageID, IsLaptop etc.).
2. **Format and Partition Disk:** Separate steps for UEFI vs BIOS. (Use variable conditions).
3. **Apply Operating System:** Apply WIM from DP (choose image index).
4. **Apply Windows Settings:** Computer name variable, local admin, timezone.
5. **Apply Network Settings:** Join domain or set static IP if needed.
6. **Apply Drivers:** Install driver package filtered by WMI.
7. **Apply Windows Update:** Use “Install Software Updates” (requires SUP).
8. **Install Applications:** In order (apps with dependencies earlier).
9. **Run Scripts / Win32 Customizations:** e.g., set registry keys, configure services.
10. **Setup Windows and ConfigMgr:** Install ccmsetup, set site code.
11. **Run SCCM Client Actions:** Policy retrieval, hardware inventory, software inventory.
12. **Final Tasks:** Cleanup, disable local admin, create final logs location, reboot to production OS.

6) Key logs and where to find them

- **smspxe.log** (on DP) — PXE boot and client PXE requests.
- **smsts.log** — *single most important* TS log.
 - In WinPE: X:\Windows\Temp\SMSTSLog\smsts.log (or X:\Windows\SMSTSLog).
 - After OS applied: C:\Windows\CCM\Logs\smsts.log or C:_SMSTaskSequence\Logs\Smstslog\smsts.log.
- **distmgr.log** (on Primary site) — distribution issues.
- **mpcontrol.log / mp_hinv.log** — MP and site communication.
- **setupact.log & setuperr.log** — Windows setup (Panther) during in-place upgrades located in C:\\$WINDOWS.~BT\Sources\Panther or C:\Windows\Panther.
- **netsetup.log** — domain join issues.
- **AppEnforce.log** — application installation during TS.

- **DISM logs** — servicing WIM or offline servicing issues.
 - **WindowsUpdate.log** — update installation problems.
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7) Common failure codes & quick remediation

- **0x80070002 / 0x80070003** — file not found / path issue: check WIM path, distribution status on DP, re-distribute content.
 - **0x80070005** — Access Denied: permissions on network share for capture, or script executed without admin.
 - **0xC1900101** — Driver issue on upgrade: uninstall / update problematic driver or BIOS. Check setuperr.log for exact driver causing failure.
 - **0x800f0922** — Not enough space or failing to access Windows Update source for components (more common in Windows Update scenarios).
 - **Task Sequence step failure** — always check smsts.log for the exact step and error message, then search for sub-step logs (e.g., AppEnforce.log for app installs).
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8) Troubleshooting checklist (stepwise)

1. **PXE failures:** Check smspxe.log on DP, verify DHCP (IP helpers vs options 66/67), ensure WDS is healthy and PXE is enabled for unknown computers if necessary.
 2. **WinPE network issues:** Use F8 command prompt, run ipconfig/ping/net use to test network and UNC share access. Confirm drivers present in boot image.
 3. **Missing TS or not offered:** Confirm TS is deployed to collection, device is in boundary group that can access content, and distribution status shows success.
 4. **smsts.log analysis:** Follow the TS step and the sub-steps in smsts.log to identify the failing action. Use the step ID to map to TS step in console.
 5. **WIM corruption:** Mount and test WIM with DISM /Get-WimInfo and DISM /Cleanup-Image or re-capture if corrupt.
 6. **App installation failures:** Check AppEnforce.log, test install manually with same command line to reproduce. Confirm content is distributed.
 7. **Upgrade rollbacks:** Collect Panther logs, identify driver/app that caused rollback, create exclusion or updated package, re-run pilot.
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9) BitLocker & security considerations

1. **Suspend BitLocker** before imaging or in-place upgrade: in TS use manage-bde - protectors -disable C: or use built-in BitLocker TS steps (pre-provision or suspend). Re-enable after completion.
 2. **Pre-provision BitLocker** in WinPE to speed encryption on large fleets (encryption in the background).
 3. **Credentials & secrets**: Do not hardcode credentials in TS; use task sequence variables or Management Point credentials and secure them. Use least privilege.
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10) Best practices & deployment tips

1. **Separate TS** for Capture, Deploy and Upgrade. Keep them simple and modular. Use nested TS for reusable blocks (e.g., Install Base Apps, Configure Security).
 2. **Driver management**: Use model-specific driver packages and WMI filtering in the TS. Avoid importing drivers into WIM except NIC/storage. Keep driver repo tidy to avoid duplicate PnP IDs.
 3. **Use pre-caching** (Peer Cache or Delivery Optimization) for large upgrades. Schedule deployments outside business hours and use maintenance windows.
 4. **Pilot first**: Validate on a small representative collection. Use stalls to catch app incompatibilities or driver failures.
 5. **Logging & Monitoring**: Configure SCCM status messages and alerts for failed TS deployments. Use real-time monitoring (SMSTS logs, DP performance) during large rollouts.
 6. **Document rollback plans**: Know how to recover devices that fail upgrade (automatic rollback, reimage, restore user state via USMT, or manual remediation).
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11) Quick commands & locations (cheat-sheet)

- **smsts.log** in WinPE: X:\Windows\Temp\SMSTSLog\smsts.log
- **capture USMT** example: scanstate \\server\store /i:migapp.xml /i:migdocs.xml /o /c
- **check DP content**: SCCM Console → Monitoring → Content Status
- **check PXE on DP**: smspxe.log on the DP (Path: C:\Windows\CCM\Logs\smspxe.log)
- **collect TS logs** on client after image: open C:\Windows\CCM\Logs\smsts.log or use Trace32 for analysis

- **mount WIM:** DISM /Mount-Wim /WimFile:C:\Images\install.wim /index:1 /MountDir:C:\mount
 - **test WIM:** DISM /Get-WimInfo /WimFile:C:\Images\install.wim
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12) Example disk partition scripts

- **UEFI GPT example (in Format and Partition Disk step):**
 - EFI: 100 MB, FAT32
 - MSR: 16 MB
 - OS (C:): Remaining space NTFS
 - Recovery: 450 MB NTFS
(Configure these values in the TS Format and Partition Disk step with VBScript/PowerShell or built-in SCCM UI.)
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